Stephen Curry: Career Shot Analysis (2015–2024)

INFSCI 1520 · Information Visualization
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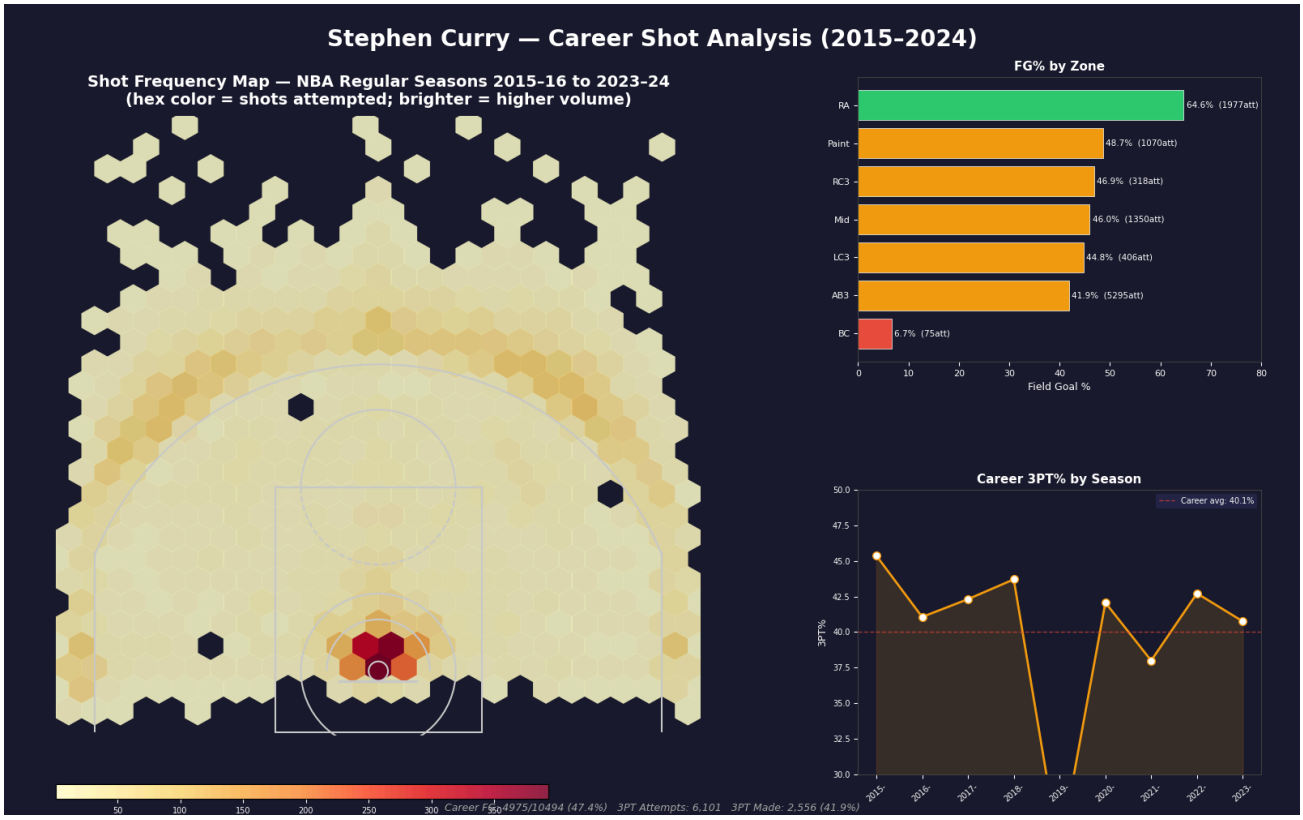

Figure 1. Stephen Curry's career shot chart covering nine NBA regular seasons (2015–16 to 2023–24). The left panel shows a hexagonal frequency map where warmer colors indicate higher shot volume. The top right panel shows field goal percentage broken down by court zone, and the bottom right panel tracks three point percentage across seasons.

Legend

Visual Element	What It Represents
Hex cell color (light to deep red)	Shot volume in that court region. Darker cells indicate more attempts were taken there.
Color scale bar (bottom of main figure)	Maps hex colors to actual attempt counts, ranging from roughly 50 to 300+ shots per cell.
Green bar in FG% chart	Zone field goal percentage at or above 50%, indicating high efficiency.
Orange bar in FG% chart	Zone field goal percentage between 40% and 50%, close to league average.
Red bar in FG% chart	Zone field goal percentage below 40%, below average efficiency.
Orange line in 3PT trend chart	Curry's three point percentage for each individual regular season.
Red dashed line in 3PT trend chart	Career three point percentage average across all nine seasons (40.1%).

Key Findings

- **Extraordinary three point volume.** Over nine seasons, Curry attempted 6,101 three pointers, which accounts for roughly 58% of all his field goal attempts. No player in NBA history has combined volume at this scale with comparable shooting efficiency.
- **Elite accuracy from distance.** A career three point percentage of 41.9% across those 6,101 attempts is well above the league wide average of around 36%. What makes this figure particularly striking is that Curry routinely shoots from well beyond the standard arc, where most players would not even consider attempting.
- **Strong finishing in the restricted area.** Despite not being a traditional slasher, Curry converts 64.6% of his shots in the restricted area on 1,977 attempts. This reflects how his off ball movement and cuts to the basket create high quality looks near the rim.
- **Mid range shooting remains effective.** Analytics era thinking has largely moved away from the mid range jumper, yet Curry shoots 46.0% from that zone on 1,350 attempts. He uses pull ups mid range looks strategically to keep defenses guessing rather than relying on them as a primary option.
- **Slight directional difference in corner threes.** The right corner three (46.9%) outperforms the left (44.8%). This kind of directional asymmetry is fairly common among right handed shooters and reflects shooting angle relative to the basket.
- **Peak efficiency in the 2015-16 season.** The three point percentage trend peaks at around 45% during the season in which Curry won the unanimous MVP award and Golden State finished with 73 wins. Even so, his career average of 40.1% holds remarkably steady across the full nine year window.
- **Shot geography confirms spacing impact.** The frequency heatmap shows dense shot clusters at distances of 25 to 30 or more feet from the basket. This spatial footprint forces defenses to extend further than normal, which in turn opens driving lanes for teammates and disrupts standard defensive rotations.

Data and Methods

Data Source. Shot level data was retrieved using the nba_api Python library, which queries the official NBA Stats endpoint at stats.nba.com. The dataset covers Stephen Curry's regular season field goal attempts across nine seasons from 2015-16 through 2023-24. Each record includes shot coordinates in tenths of feet from the basket, shot type (two pointer or three pointer), the NBA-defined shot zone (Restricted Area, Paint, Mid Range, Corner Three, Above-the-Break Three, Backcourt), and the outcome of each attempt. The total dataset contains 10,494 field goal attempts.

Analytical Procedures. All data processing and visualization were done in Python. Shot coordinates were grouped into hexagonal bins using Matplotlib's hexbin function with a gridsize of 25, plotted over the half court coordinate space. Zone level field goal percentages were computed by aggregating made and attempted shots per NBA shot zone category, and zones with fewer than 50 attempts were excluded to avoid misleading small sample results. Season level three point percentages were derived by filtering for three point attempts and grouping by season. Court geometry including the paint, three point arc, and restricted area was drawn using Matplotlib patch objects scaled to match NBA coordinate conventions.

Libraries used: nba_api 1.x, pandas, matplotlib

Significance

Stephen Curry is probably the most talked-about shooter in NBA history, but most of that conversation stays at the level of highlights and stats lines. This project tries to ground it in something more concrete by looking at where he actually shoots, how often, and how well.

The most striking thing the data shows is just how far out he operates. The heatmap makes clear that a huge portion of his attempts come from distances where most players would not even set their feet. That spatial pressure on defenses is real, and it helps explain a lot of what Golden State was able to do offensively during this period.

The zone accuracy chart adds another layer. Curry is not just a three point specialist, he shoots above league average from nearly every zone on the floor, including mid range and the restricted area. That kind of all court efficiency is rare and does not always come through in the headline numbers.

On the methods side, everything here uses publicly available data pulled directly from the NBA Stats API. The goal was to show that this kind of fancy analysis does not require proprietary tools or a front office budget — just Python and a bit of time.

GitHub Repository

All source code, data fetching scripts, and this report are archived at:
https://github.com/huntingdrew/INF1520-curry-shot-analysis
The repository contains curry_shots.py (data fetching and visualization), the main figure image, and the final report.

Data source: NBA Stats API via nba_api · Visualization: Python / Matplotlib